

# Modelling Housing Affordability in England Using Regression and Statistical Learning Methods

MATH3092: Mathematics Project

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## Abstract

This project models housing affordability across nine English regions over 2004 to 2022 using panel data from the Office for National Statistics. Housing affordability in England has worsened significantly over this period, with the national mean house price to earnings ratio rising from 6.32 in 2004 to 8.48 in 2022. The analysis is carried out in two stages: first, a fixed effects regression model is built to identify which variables drive affordability, with formal diagnostic testing used to justify model choices. Second, several models are compared on their ability to predict out-of-sample data. The regression results show that net housing additions, mortgage rates, population growth and regional productivity (GVA per hour) are the key determinants of within-region affordability. When comparing predictive performance, OLS outperforms Ridge, LASSO and Random Forest, suggesting that a simple well-specified model works better than more complex methods for this type of data.

## 1 Introduction

Housing affordability in England has worsened over recent decades, with house prices increasing faster than earnings across all regions. The affordability ratio, which is calculated as median house prices divided by median workplace-based earnings, increased from a national mean of 6.32 in 2004 to 8.48 in 2022 (ONS, 2024). Regional differences have widened significantly throughout the sample period. For example, London’s affordability ratio reached 12.40 in 2022, placing it in the “Impossibly Unaffordable” category according to the Demographia classification (Cox, 2024), while the North East remained the most affordable region at 5.25. These persistent differences across regions reflect local planning constraints, geographic factors and labour market conditions, which motivate the use of region-specific controls in the modelling framework.

Regional affordability data are structured as panel data indexed by region and year. This structure motivates a two-stage framework: first, a fixed effects panel regression is used to identify the key macroeconomic determinants of housing affordability, and second, statistical learning methods including Ridge regression, LASSO and Random Forest are tested to assess whether regularisation improves out-of-sample predictive performance relative to OLS (James et al., 2021).

The remainder of this report is structured as follows. Section 2 reviews the relevant literature. Section 3 describes the data and variables. Section 4 develops the mathematical framework. Section 5 presents the empirical results. Section 6 concludes.

## 2 Background

### 2.1 Measuring Housing Affordability

Housing affordability is most commonly measured using the house price-to-income ratio, defined as the ratio of median house prices to median household or workplace-based earnings within a given area and time period (Hilber and Schöni, 2016). This measure provides a simple and consistent way to describe how housing costs compare to income, and is used extensively in academic research and policy analysis.

The Demographia International Housing Affordability Report (Cox, 2024) classifies markets according to this ratio as shown in Table 1.

Table 1: Demographia Housing Affordability Classifications

| Rating                  | Affordability Ratio |
|-------------------------|---------------------|
| Affordable              | 3.0 and under       |
| Moderately Unaffordable | 3.1 to 4.0          |
| Seriously Unaffordable  | 4.1 to 5.0          |
| Severely Unaffordable   | 5.1 to 8.9          |
| Impossibly Unaffordable | 9.0 and over        |

*Source: Cox (2024). Affordability ratio defined as median house price divided by median household income.*

By this classification, the majority of English regions have moved from the “Seriously Unaffordable” into the “Severely Unaffordable” category over the sample period, with London and the South East approaching or exceeding the “Impossibly Unaffordable” threshold.

### 2.2 Determinants of Housing Affordability

Housing affordability is driven by both demand-side and supply-side factors. On the demand side, labour market conditions such as unemployment, population growth, mortgage rates and regional productivity are frequently cited determinants (Pintér, 2018). On the supply side, net housing additions are expected to improve affordability, though the empirical evidence on the size of this effect is mixed, particularly over short time periods (Hilber and Vermeulen, 2016; Fingleton et al., 2019). In addition, persistent regional differences that do not change over time, such as local planning restrictions and geographic constraints, motivate the use of region-specific intercepts in the regression framework.

## 2.3 Panel Data Approaches

Panel models are useful in this setting because they can simultaneously account for cross-sectional differences and time trends across regions. Fixed effects specifications are particularly appropriate when regional characteristics are likely to be correlated with the included predictors, which is plausible in this case given the differences in regional labour markets and productivity. Alongside classical regression, statistical learning methods including Ridge regression and LASSO are used to handle correlated predictors and are evaluated using out-of-sample prediction rather than in-sample fit, which gives a more honest assessment of model performance (James et al., 2021).

## 3 Data and Variables

### 3.1 Data Sources

All data are drawn from the Office for National Statistics (ONS), the Bank of England, and the Ministry of Housing, Communities and Local Government (MHCLG). The unit of observation is a region–year pair, with the sample covering nine English regions over the period 2004–2022, yielding a perfectly balanced panel of  $N = 9 \times T = 9 \times 19 = 171$  observations.

The nine English regions are: North East, North West, Yorkshire and The Humber, East Midlands, West Midlands, East of England, London, South East, and South West. The sample period is determined by the intersection of availability across all variables. Table 2 summarises the raw coverage of each series prior to merging.

Table 2: Data Coverage by Variable

| Variable              | Raw coverage | Level    | Source          |
|-----------------------|--------------|----------|-----------------|
| Affordability ratio   | 1997–2024    | Region   | ONS             |
| Unemployment rate     | 2004–2023    | Region   | ONS             |
| Net housing additions | 2001–2024    | Region   | ONS/MHCLG       |
| Mortgage rate         | 2004–2022    | National | Bank of England |
| Population            | 1991–2024    | Region   | ONS             |
| Inflation (CPIH)      | 1989–2025    | National | ONS             |
| GVA per hour          | 2004–2023    | Region   | ONS             |

*The estimation sample 2004–2022 is determined by the binding constraint of the mortgage rate series, which is available only to 2022. All variables are merged on region and year; rows with any missing predictor are dropped, yielding the balanced panel of 171 observations.*

The lower bound of 2004 reflects the start of the regional unemployment and GVA series. The upper bound of 2022 is imposed by the mortgage rate data, which is the

binding constraint: extending to 2023 would require dropping mortgage rate from the model or accepting an unbalanced panel, both of which would compromise comparability across specifications.

### 3.2 Response Variable

The response variable is the house price-to-workplace-based earnings ratio, denoted  $y_{i,t}$ , which compares median house prices with median annual workplace-based earnings within each region  $i$  at time  $t$ . This ratio is sourced from the ONS housing affordability publication (ONS, 2024) and provides a consistent measure of affordability that is comparable across regions and over time.

Since the affordability ratio incorporates median income in the denominator, care is taken to avoid including income-proxying variables as predictors, as these would create a mechanical relationship with the response. All predictor variables are therefore selected to capture supply-side constraints and demand-side pressures independently of income levels.

### 3.3 Explanatory Variables

The explanatory variables are described in Table 3. They are selected to capture key demand-side and supply-side determinants of regional affordability, subject to data availability.

Table 3: Variable Descriptions and Sources

| Variable         | Description                               | Level    | Source          |
|------------------|-------------------------------------------|----------|-----------------|
| UnemploymentRate | Regional unemployment rate (%)            | Region   | ONS             |
| NetHousing       | Net housing additions per 1,000 dwellings | Region   | ONS/MHCLG       |
| MortgageRate     | Average fixed-rate mortgage rate (%)      | National | Bank of England |
| PopGrowth        | Annual population growth rate (%)         | Region   | ONS             |
| Inflation        | CPIH annual inflation rate (%)            | National | ONS             |
| GVA              | GVA per hour worked (index)               | Region   | ONS             |

*MortgageRate is computed as the annual average of monthly fixed-rate mortgage observations. PopGrowth is computed as the year-on-year percentage change in regional population.*

Two variables, MortgageRate and Inflation have no regional variation, as they are national series. In a fixed effects model, these variables are identified solely from their within-year time-series variation and are therefore harder to distinguish from the common time trend. This is acknowledged as a limitation of the current specification.

### 3.4 Descriptive Statistics

Table 4 presents summary statistics for the response variable and all predictors over the estimation sample.

Table 4: Descriptive Statistics (2004–2022,  $N = 171$ )

| Variable           | Mean  | Std Dev | Min  | Median | Max  |
|--------------------|-------|---------|------|--------|------|
| AffordabilityRatio | 7.20  | 1.85    | 4.8  | 6.83   | 12.9 |
| UnemploymentRate   | 5.69  | 1.85    | 2.6  | 5.30   | 10.6 |
| NetHousing         | 8.17  | 2.18    | 3.0  | 8.00   | 13.0 |
| MortgageRate       | 3.87  | 1.35    | 2.0  | 4.17   | 5.6  |
| PopGrowth          | 0.68  | 0.35    | -0.7 | 0.67   | 1.8  |
| Inflation          | 2.42  | 1.53    | 0.4  | 2.30   | 7.9  |
| GVA                | 30.85 | 6.32    | 22.0 | 29.13  | 52.8 |

*Affordability ratio is the house price-to-workplace-based earnings ratio. All other variables are defined in Table 3.*

### 3.5 Preliminary Observations

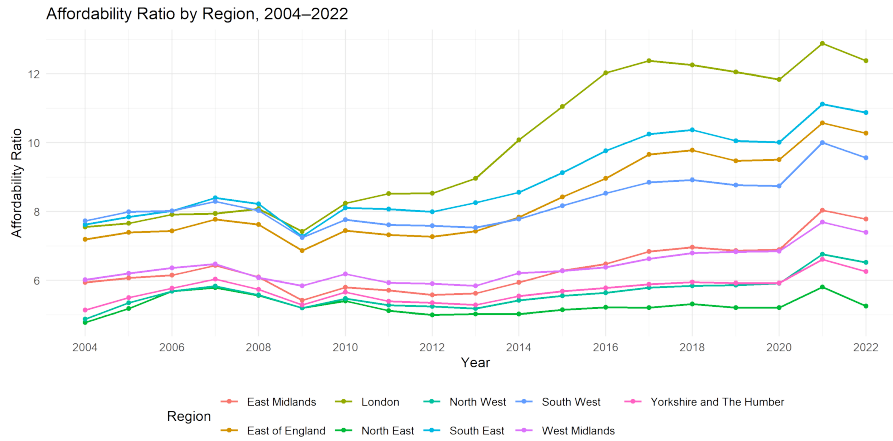


Figure 1: Affordability Ratio by Region, 2004–2022

Figure 1 illustrates the evolution of affordability ratios across the nine English regions over the sample period 2004–2022. The national mean affordability ratio rose from 6.32 in 2004 to a peak of 8.83 in 2021, before falling slightly to 8.48 in 2022, indicating that housing has become substantially less affordable relative to earnings over this period. The range across regions widened considerably: in 2004, ratios spanned from 4.78 (North East) to 7.73 (South West); by 2022, this range had expanded to 5.25 (North East) to 12.40 (London), reflecting a divergence in affordability outcomes between northern and southern regions.

Table 5 presents the affordability ratio and Demographia classification for each region in 2004 and 2022.

Table 5: Regional Affordability Ratios and Demographia Classifications, 2004 and 2022

| 2004                   |                    | 2022                   |                     |
|------------------------|--------------------|------------------------|---------------------|
| Region                 | Ratio              | Region                 | Ratio               |
| North East             | 4.78 <sup>SI</sup> | North East             | 5.25 <sup>SV</sup>  |
| North West             | 4.87 <sup>SI</sup> | Yorkshire & The Humber | 6.26 <sup>SV</sup>  |
| Yorkshire & The Humber | 5.14 <sup>SV</sup> | North West             | 6.52 <sup>SV</sup>  |
| East Midlands          | 5.94 <sup>SV</sup> | West Midlands          | 7.39 <sup>SV</sup>  |
| West Midlands          | 6.02 <sup>SV</sup> | East Midlands          | 7.78 <sup>SV</sup>  |
| East of England        | 7.19 <sup>SV</sup> | South West             | 9.56 <sup>IM</sup>  |
| London                 | 7.55 <sup>SV</sup> | East of England        | 10.30 <sup>IM</sup> |
| South East             | 7.62 <sup>SV</sup> | South East             | 10.90 <sup>IM</sup> |
| South West             | 7.73 <sup>SV</sup> | London                 | 12.40 <sup>IM</sup> |
| <b>National mean</b>   | <b>6.32</b>        | <b>National mean</b>   | <b>8.48</b>         |

*Superscripts denote Demographia classifications: SI = Seriously Unaffordable (4.1–5.0), SV = Severely Unaffordable (5.1–8.9), IM = Impossibly Unaffordable ( $\geq 9.0$ ). Regions ranked by affordability ratio within each year. Source: ONS; Demographia classifications from Cox (2024).*

Several patterns are evident from Table 5. In 2004, all nine regions fell within either the Seriously or Severely Unaffordable categories. By 2022, no region remained Seriously Unaffordable, and four regions which are South West, East of England, South East, and London had crossed into the Impossibly Unaffordable category ( $\geq 9.0$ ). London’s ratio of 12.40 in 2022 is particularly striking, representing an increase of 64% from its 2004 level of 7.55. The North East, by contrast, remained the most affordable region throughout, with a ratio of 5.25 in 2022, though even this represents a deterioration from 4.78 in 2004.

The persistence of the regional ordering with northern regions consistently more affordable than southern regions motivates the fixed effects specification, which controls for these time-invariant structural differences through region-specific intercepts.

## 4 Mathematical and Statistical Framework

### 4.1 Notation

Let  $y_{i,t}$  denote the log-transformed housing affordability ratio in region  $i = 1, \dots, N$  at time  $t = 1, \dots, T$ , where  $N = 9$  and  $T = 19$ . The log transformation is applied throughout to reduce skewness and allow coefficients to be interpreted as approximate

percentage effects. Let  $\mathbf{x}_{i,t} \in \mathbb{R}^p$  denote the vector of time-varying predictors and  $\alpha_i$  denote unobserved, time-invariant regional characteristics.

## 4.2 Baseline Regression Models

The analysis begins with a sequence of progressively richer specifications that serve as benchmarks for the panel models developed in Section 4.3.

**Equation 1: Time trend model.** The simplest specification captures the long-run upward movement in affordability common across all regions:

$$y_{i,t} = \beta_0 + \beta_1 t + \varepsilon_{i,t}, \quad (1)$$

where  $t$  denotes the year and  $\varepsilon_{i,t}$  is an idiosyncratic error term. This model serves as a minimal benchmark that abstracts from all cross-sectional variation.

**Equation 2: Region-specific intercepts (Least Square Dummy Variables).** To account for persistent regional differences in baseline affordability levels, the LSDV approach adds a dummy variable for each region, allowing each region to have its own baseline level of affordability. This is captured through the region-specific intercept  $\alpha_i$ , which is introduced alongside the full vector of macroeconomic predictors  $\mathbf{x}_{i,t} \in \mathbb{R}^6$ , containing the regional unemployment rate, net housing additions, mortgage rate, population growth rate, inflation, and GVA per hour:

$$y_{i,t} = \alpha_i + \mathbf{x}_{i,t}^\top \boldsymbol{\beta} + \varepsilon_{i,t}. \quad (2)$$

In the LSDV approach,  $\alpha_i$  is implemented through  $N - 1$  region indicator variables. The within estimator of Section 4.3 is algebraically equivalent but eliminates  $\alpha_i$  through demeaning rather than explicit dummy variables.

## 4.3 Fixed Effects Estimation

### 4.3.1 The Within Estimator

The full panel model is specified as in Equation (2), where  $\varepsilon_{i,t}$  is assumed to satisfy  $\mathbb{E}[\varepsilon_{i,t} \mid \mathbf{x}_{i,t}, \alpha_i] = 0$  under fixed effects.

The key idea behind the within estimator is to remove the unobserved regional effect  $\alpha_i$  by subtracting each region's time average from its observed values. To see how this works, consider first a simple model with a single explanatory variable:

$$y_{i,t} = \beta_1 x_{i,t} + \alpha_i + \varepsilon_{i,t}. \quad (3)$$

Averaging equation (3) over time for each region  $i$  gives:

$$\bar{y}_i = \beta_1 \bar{x}_i + \alpha_i + \bar{\varepsilon}_i, \quad (4)$$

where  $\bar{y}_i = T^{-1} \sum_{t=1}^T y_{i,t}$  and similarly for  $\bar{x}_i$  and  $\bar{\varepsilon}_i$ . Since  $\alpha_i$  is fixed over time, it appears in both (3) and (4). Subtracting (4) from (3) eliminates  $\alpha_i$ :

$$y_{i,t} - \bar{y}_i = \beta_1 (x_{i,t} - \bar{x}_i) + (\varepsilon_{i,t} - \bar{\varepsilon}_i), \quad t = 1, 2, \dots, T. \quad (5)$$

Writing  $\tilde{y}_{i,t} = y_{i,t} - \bar{y}_i$  and  $\tilde{x}_{i,t} = x_{i,t} - \bar{x}_i$  for the demeaned variables, this simplifies to:

$$\tilde{y}_{i,t} = \beta_1 \tilde{x}_{i,t} + \tilde{\varepsilon}_{i,t}. \quad (6)$$

The unobserved effect  $\alpha_i$  has been eliminated entirely. OLS applied to (6) on the demeaned data gives an unbiased estimate of  $\beta_1$  (Wooldridge, 2019).

Extending this to the case of multiple predictors  $\mathbf{x}_{i,t} \in \mathbb{R}^p$ , the demeaning operation is conveniently expressed using the demeaning operator for region  $i$ :

$$M_i = I_T - \frac{1}{T} \iota_T \iota_T^\top, \quad (7)$$

where  $I_T$  is the  $T \times T$  identity matrix and  $\iota_T$  is a  $T$ -vector of ones. Applying  $M_i$  to both sides of (2) yields the within-transformed model:

$$\tilde{y}_{i,t} = \tilde{\mathbf{x}}_{i,t}^\top \boldsymbol{\beta} + \tilde{\varepsilon}_{i,t}, \quad (8)$$

where  $\tilde{y}_{i,t} = y_{i,t} - \bar{y}_i$ ,  $\tilde{\mathbf{x}}_{i,t} = \mathbf{x}_{i,t} - \bar{\mathbf{x}}_i$ , and  $\tilde{\varepsilon}_{i,t} = \varepsilon_{i,t} - \bar{\varepsilon}_i$  denote deviations from regional means. The regional effects  $\alpha_i$  are eliminated because  $M_i \alpha_i \iota_T = 0$ .

The within estimator is then obtained by applying OLS to (8):

$$\hat{\boldsymbol{\beta}}_{FE} = \left( \tilde{X}^\top \tilde{X} \right)^{-1} \tilde{X}^\top \tilde{y}, \quad (9)$$

where  $\tilde{X}$  and  $\tilde{y}$  collect the demeaned observations. By the Frisch–Waugh–Lovell theorem, the within estimator (9) is algebraically equivalent to the LSDV estimator that explicitly includes region dummy variables. The slope coefficients are therefore identical across both approaches, though the within estimator is preferred for its computational efficiency and cleaner output.

Note that demeaning reduces the degrees of freedom: for each region  $i$ , one degree of freedom is lost because the demeaned errors  $\tilde{\varepsilon}_{i,t}$  sum to zero across  $t$ . The appropriate degrees of freedom for the within estimator is therefore  $NT - N - k$ , where  $k$  is the number of predictors (Wooldridge, 2019).

Under the standard fixed effects assumptions strict exogeneity, no multicollinearity

in  $\tilde{X}$ , and homoskedastic, serially uncorrelated errors, the within estimator is unbiased, consistent, and efficient within the class of linear estimators that remove individual effects (Baltagi, 2021).

### 4.3.2 The Random Effects Estimator

The random effects (RE) estimator imposes the additional assumption that the unobserved regional effects are uncorrelated with the regressors:

$$\text{Cov}(\alpha_i, \mathbf{x}_{i,t}) = 0 \quad \forall i, t. \quad (10)$$

Under this assumption, the RE estimator is more efficient than the FE estimator because it exploits both within-region and between-region variation. However, if assumption (10) fails since it is plausible when unobserved regional characteristics such as planning constraints or geographic amenities are correlated with productivity or labour market conditions then, the RE estimator is inconsistent, while the FE estimator remains consistent.

### 4.3.3 The Hausman Test

The choice between FE and RE is guided by the Hausman test (Hausman, 1978), which tests  $H_0 : \text{Cov}(\alpha_i, \mathbf{x}_{i,t}) = 0$  against the alternative that this covariance is non-zero. Under  $H_0$ , both estimators are consistent but RE is efficient; under  $H_1$ , only FE is consistent. The test statistic is:

$$H = \left( \hat{\beta}_{FE} - \hat{\beta}_{RE} \right)^\top \left[ \widehat{\text{Var}}(\hat{\beta}_{FE}) - \widehat{\text{Var}}(\hat{\beta}_{RE}) \right]^{-1} \left( \hat{\beta}_{FE} - \hat{\beta}_{RE} \right) \xrightarrow{d} \chi^2(k), \quad (11)$$

where  $k$  is the number of time-varying regressors.

## 4.4 Multicollinearity and Variable Selection

### 4.4.1 Variance Inflation Factors

Multicollinearity among regressors inflates the variance of OLS estimates without introducing bias, making it difficult to isolate the individual effects of correlated predictors. The Variance Inflation Factor (VIF) for predictor  $j$  is defined as:

$$\text{VIF}_j = \frac{1}{1 - R_j^2}, \quad (12)$$

where  $R_j^2$  is the coefficient of determination from regressing predictor  $j$  on all other predictors. Values exceeding 10 are conventionally regarded as indicative of problematic multicollinearity.

#### 4.4.2 Backward Elimination

Variable selection proceeds by backward elimination: beginning with the full model, predictors are removed one at a time in order of decreasing  $p$ -value, provided that removal does not materially reduce the adjusted coefficient of determination  $\bar{R}^2$ . Each elimination step is formalised using a Wald test, which tests the null hypothesis that the coefficients of the removed predictors are jointly zero. Under the null hypothesis, the test statistic follows a  $\chi^2$  distribution with degrees of freedom equal to the number of restrictions.

### 4.5 Inference Under Serial Correlation and Cross-Sectional Dependence

Panel data frequently exhibit serial correlation within cross-sectional units and cross-sectional dependence across units. Both features invalidate the standard OLS standard errors, which assume independent and identically distributed errors.

Serial correlation is tested using the Breusch–Godfrey test for panels (Baltagi, 2021), and cross-sectional dependence using the Pesaran CD test (Pesaran, 2021). Given significant evidence of both, Driscoll–Kraay standard errors are used throughout (Driscoll and Kraay, 1998; Hoechle, 2007). These are consistent under arbitrary heteroskedasticity, serial correlation up to some lag, and cross-sectional dependence, making them appropriate for regional panel data subject to common national shocks.

### 4.6 Statistical Learning Methods

#### 4.6.1 Ordinary Least Squares

Let  $\mathbf{y} \in \mathbb{R}^n$  denote the vector of log-transformed affordability ratios and  $X \in \mathbb{R}^{n \times p}$  the design matrix of predictors. The OLS estimator minimises the residual sum of squares:

$$\hat{\boldsymbol{\beta}}_{OLS} = \arg \min_{\boldsymbol{\beta}} \frac{1}{n} \|\mathbf{y} - X\boldsymbol{\beta}\|_2^2 = (X^\top X)^{-1} X^\top \mathbf{y}, \quad (13)$$

provided  $X^\top X$  is non-singular. OLS is unbiased and, under the Gauss–Markov assumptions, efficient within the class of linear unbiased estimators. However, it can be unstable when predictors are correlated, motivating the use of shrinkage methods.

#### 4.6.2 Ridge Regression

Ridge regression (Hoerl and Kennard, 1970; James et al., 2021) extends OLS by adding a penalty on the squared Euclidean norm of the coefficient vector:

$$\hat{\boldsymbol{\beta}}_{ridge}(\lambda) = \arg \min_{\boldsymbol{\beta}} \frac{1}{n} \|\mathbf{y} - X\boldsymbol{\beta}\|_2^2 + \lambda \|\boldsymbol{\beta}\|_2^2, \quad (14)$$

for a penalty parameter  $\lambda \geq 0$ . This is equivalent to the constrained optimisation problem:

$$\min_{\boldsymbol{\beta}} \|\mathbf{y} - X\boldsymbol{\beta}\|_2^2 \quad \text{subject to} \quad \|\boldsymbol{\beta}\|_2^2 \leq t, \quad (15)$$

where the constraint set is a ball in  $\mathbb{R}^p$ . Unlike the LASSO constraint set (see below), this ball is strictly convex and smooth, so the solution path is continuous in  $\lambda$  and never exactly zero.

The Ridge estimator has a closed-form solution:

$$\hat{\boldsymbol{\beta}}_{\text{ridge}}(\lambda) = (X^\top X + \lambda I_p)^{-1} X^\top \mathbf{y}, \quad (16)$$

which recovers OLS when  $\lambda = 0$ . Adding  $\lambda I_p$  to  $X^\top X$  guarantees positive definiteness even when  $X^\top X$  is singular or poorly conditioned, making Ridge well-defined for any  $\lambda > 0$ .

**Bias–variance decomposition.** Ridge introduces a bias in exchange for a reduction in variance. The bias and variance of the Ridge estimator are:

$$\text{Bias}(\hat{\boldsymbol{\beta}}_{\text{ridge}}) = -\lambda (X^\top X + \lambda I_p)^{-1} \boldsymbol{\beta}, \quad (17)$$

$$\text{Var}(\hat{\boldsymbol{\beta}}_{\text{ridge}}) = \sigma^2 (X^\top X + \lambda I_p)^{-1} X^\top X (X^\top X + \lambda I_p)^{-1}. \quad (18)$$

As  $\lambda$  increases, the variance decreases monotonically while the bias increases. The optimal  $\lambda$  balances this trade-off to minimise mean squared prediction error. In settings where  $X^\top X$  is already well-conditioned as when  $p$  is small relative to  $n$ , this variance reduction is modest, and the bias cost may outweigh the benefit.

### 4.6.3 The LASSO

The LASSO (Tibshirani, 1996; James et al., 2021) replaces the squared penalty with an  $\ell_1$  penalty:

$$\hat{\boldsymbol{\beta}}_{\text{lasso}}(\lambda) = \arg \min_{\boldsymbol{\beta}} \frac{1}{n} \|\mathbf{y} - X\boldsymbol{\beta}\|_2^2 + \lambda \|\boldsymbol{\beta}\|_1, \quad (19)$$

which is equivalent to:

$$\min_{\boldsymbol{\beta}} \|\mathbf{y} - X\boldsymbol{\beta}\|_2^2 \quad \text{subject to} \quad \|\boldsymbol{\beta}\|_1 \leq t. \quad (20)$$

The  $\ell_1$  constraint set is a cross-polytope whose corners lie on the coordinate axes. Because the loss function’s level sets are ellipsoidal and the constraint set has corners, the optimal solution frequently occurs at a corner where some coefficients are exactly zero. This geometric property gives the LASSO its variable selection capability, which Ridge with its smooth spherical constraint cannot achieve.

The LASSO solution has no closed form but is computed efficiently using coordinate descent algorithms.

#### 4.6.4 Lambda Selection via Cross-Validation

For all shrinkage methods, the penalty parameter  $\lambda$  is selected by  $k$ -fold cross-validation. The training set is partitioned into  $k$  folds  $\{V_1, \dots, V_k\}$ , and  $\lambda$  is chosen to minimise the average held-out prediction error:

$$\hat{\lambda} = \arg \min_{\lambda} \frac{1}{k} \sum_{j=1}^k \frac{1}{|V_j|} \sum_{i \in V_j} \left( y_i - \hat{f}_{\lambda}^{-j}(\mathbf{x}_i) \right)^2, \quad (21)$$

where  $\hat{f}_{\lambda}^{-j}$  denotes the model trained on all folds except  $V_j$ .

Two values of  $\lambda$  are reported:  $\lambda_{\min}$ , which minimises the cross-validated error directly, and  $\lambda_{1se}$ , the largest  $\lambda$  within one standard error of  $\lambda_{\min}$ . The latter corresponds to a more parsimonious model that sacrifices a small amount of predictive accuracy for additional regularisation.

#### 4.6.5 Random Forest

Random Forest (Breiman, 2001) is a non-parametric ensemble method that aggregates predictions from  $B$  decision trees fitted on bootstrap samples of the training data, giving:

$$\hat{f}(x) = \frac{1}{B} \sum_{b=1}^B T_b(x) \quad (22)$$

At each split, only a random subset of  $m$  predictors is considered, which decorrelates the trees and reduces ensemble variance. The tuning parameter  $m$  is selected by minimising the out-of-bag (OOB) error via `tuneRF` in R, with  $B = 500$  trees grown. Random Forest is included as a nonparametric benchmark as it places no assumptions on the functional form of the relationship between predictors and the response.

#### 4.6.6 Model Evaluation

Predictive performance is assessed using the root mean squared error (RMSE) on a time-based holdout sample:

$$\text{RMSE} = \sqrt{\frac{1}{n_{test}} \sum_{i,t \in \mathcal{T}_{test}} (y_{i,t} - \hat{y}_{i,t})^2}, \quad (23)$$

where  $\mathcal{T}_{test}$  denotes the set of region-year pairs in the test period ( $t > 2018$ ). A temporal split is used rather than a random split to respect the panel structure of the data: random

splitting would allow future observations to appear in the training set, producing an optimistic and invalid estimate of out-of-sample performance.

## 5 Results

### 5.1 Diagnostic Testing and Model Selection

#### 5.1.1 Justifying the Fixed Effects Specification

Table 6 reports the results of the panel diagnostic tests used to justify the fixed effects specification.

Table 6: Panel Diagnostic Test Results

| Test                            | Null Hypothesis       | Statistic         | d.f.     | p-value   |
|---------------------------------|-----------------------|-------------------|----------|-----------|
| F-test (FE vs Pooled OLS)       | No individual effects | $F = 60.83$       | (8, 156) | $< 0.001$ |
| Breusch–Pagan LM (RE vs Pooled) | No panel effects      | $\chi^2 = 420.35$ | 1        | $< 0.001$ |
| Hausman (FE vs RE)              | RE is consistent      | $\chi^2 = 3.43$   | 6        | 0.753     |
| Breusch–Godfrey (serial corr.)  | No serial correlation | $\chi^2 = 102.23$ | 19       | $< 0.001$ |
| Pesaran CD (cross-sect. dep.)   | No cross-sect. dep.   | $z = 6.47$        | —        | $< 0.001$ |

The F-test strongly rejects the null hypothesis of no individual effects ( $F = 60.83$ ,  $p < 0.001$ ), confirming that regional fixed effects are jointly highly significant and that pooled OLS is inadequate. The Breusch–Pagan test likewise rejects pooled OLS in favour of a panel specification ( $\chi^2 = 420.35$ ,  $p < 0.001$ ), confirming the presence of substantial unobserved heterogeneity across regions. The Hausman test fails to reject the consistency of the random effects estimator ( $\chi^2 = 3.43$ ,  $p = 0.753$ ). However, this result should be interpreted cautiously for two reasons. First, the test has low power with only  $N = 9$  cross-sectional units (Hausman, 1978; Baltagi, 2021). Second, and more fundamentally, there are strong theoretical grounds for preferring fixed effects in this setting: unobserved regional characteristics such as local planning constraints, geographic amenities, and historical development patterns are plausibly correlated with the included regressors (particularly GVA and unemployment), violating the key assumption (10) underlying the random effects estimator. The fixed effects specification is therefore adopted throughout on both statistical and theoretical grounds. The Breusch–Godfrey test detects strong serial correlation in the idiosyncratic errors ( $\chi^2 = 102.23$ ,  $p < 0.001$ ), consistent with the gradual, trend-driven nature of regional housing markets. The Pesaran CD test reveals statistically significant cross-sectional dependence ( $z = 6.47$ ,  $p < 0.001$ ), indicating that regional affordability outcomes co-move in response to common national shocks such as changes in monetary policy or national planning legislation. Together, these findings mo-

tivate the use of Driscoll–Kraay standard errors for all subsequent inference, which are robust to both sources of error correlation.

### 5.1.2 Multicollinearity

Table 7 reports Variance Inflation Factors for the full model specification including the linear year trend.

Table 7: Variance Inflation Factors

| Variable         | (1) With Year | (2) Year Dropped | (3) Within-Demeaned |
|------------------|---------------|------------------|---------------------|
| Year             | 68.19         | —                | —                   |
| GVA              | 71.99         | 24.74            | 10.70               |
| MortgageRate     | 34.92         | 12.48            | 12.48               |
| UnemploymentRate | 9.99          | 7.30             | 4.95                |
| NetHousing       | 5.64          | 5.47             | 3.41                |
| PopGrowth        | 2.15          | 1.85             | 1.29                |
| Inflation        | 2.04          | 2.02             | 2.02                |

*(1) Full model including linear year trend; (2) Year removed after identifying Year/GVA collinearity; (3) Within-demeaned data as seen by the FE estimator after removing region means. GVIF values for region dummies are not reported as they mechanically inflate under any fixed effects specification. The marginal exceedance of GVA and MortgageRate in column (3) reflects their strong negative correlation ( $r = -0.90$ ) over time, which is addressed by the removal of MortgageRate during backward elimination.*

The linear year trend and GVA per hour exhibit near-identical GVIFs of 68.19 and 71.99 respectively, indicating severe multicollinearity. This arises because both variables capture the upward trend in regional economic conditions over the sample period: GVA per hour has increased consistently across all regions since 2004, generating near-perfect collinearity with the year counter in the LSDV design matrix, i.e.  $X^\top X$  approaches singularity. To resolve this, the year trend is dropped from the preferred specification. GVA per hour is retained because it contains the time trend while additionally capturing cross-regional variation in productivity which is a richer source of identification than year alone. Following this adjustment, all remaining predictors have GVIF below 10 with the exception of MortgageRate (GVIF = 12.48) and GVA (GVIF = 24.74), which reflect between-region variation rather than within-region collinearity. Column (3) of Table 7 reports GVIFs on within-demeaned data, replicating exactly what the FE estimator sees after removing region means. Most variables fall comfortably below 10, though GVA (10.70) and MortgageRate (12.48) remain marginally elevated due to their strong negative correlation over time ( $r = -0.90$ ). MortgageRate is subsequently removed during backward elimination, resolving this residual concern.

## 5.2 Fixed Effects Regression Results

### 5.2.1 Specification Comparison

Table 8 reports the fixed effects estimates across four progressive specifications, with Driscoll–Kraay standard errors throughout.

Table 8: Fixed Effects Regression Results

|                  | (1)                   | (2)                   | (3)                   |
|------------------|-----------------------|-----------------------|-----------------------|
|                  | Time trend            | Full FE               | Final FE              |
| Year             | 0.0145***<br>(0.0026) | —                     | —                     |
| UnemploymentRate | —                     | −0.0087<br>(0.0077)   | —                     |
| NetHousing       | —                     | 0.0222***<br>(0.0032) | 0.0273***<br>(0.0019) |
| MortgageRate     | —                     | 0.0289<br>(0.0162)    | 0.0250*<br>(0.0099)   |
| PopGrowth        | —                     | −0.0673*<br>(0.0268)  | −0.0650**<br>(0.0204) |
| Inflation        | —                     | 0.0007<br>(0.0068)    | —                     |
| GVA              | —                     | 0.0252***<br>(0.0049) | 0.0250***<br>(0.0024) |
| Region FE        | Yes                   | Yes                   | Yes                   |
| Observations     | 171                   | 171                   | 171                   |
| $R^2$            | 0.895                 | 0.953                 | 0.952                 |
| Within $R^2$     | 0.513                 | 0.782                 | 0.779                 |

*Driscoll–Kraay standard errors in parentheses.  $p < 0.10$ ,  $*p < 0.05$ ,  $**p < 0.01$ ,  $***p < 0.001$ . All specifications include region fixed effects estimated via the within estimator.*

### 5.2.2 Backward Elimination

Starting from the full specification, backward elimination proceeds by removing predictors in order of decreasing  $p$ -value. UnemploymentRate is removed first ( $p = 0.155$ ), followed by Inflation ( $p = 0.512$ ). Table 9 records the adjusted  $R^2$  at each step.

Table 9: Backward Elimination: Adjusted  $R^2$  Comparison

| Model          | Variables removed           | Adj. $R^2$ |
|----------------|-----------------------------|------------|
| Full model     | —                           | 0.7627     |
| Step 1         | UnemploymentRate            | 0.7611     |
| Step 2 (Final) | UnemploymentRate, Inflation | 0.7620     |

The Wald test confirms that the reductions in model fit at each step are not statistically significant ( $\chi^2 = 2.48$ ,  $p = 0.290$ ), formally justifying the elimination of these predictors. The final specification therefore retains MortgageRate, PopGrowth, NetHousing, and GVA as regressors alongside region fixed effects.

### 5.2.3 Interpretation of Final Model

The final fixed effects estimates (column 3 of Table 8) are interpreted as follows. All coefficients represent within-region effects and the estimated relationship between changes in each predictor and changes in the log affordability ratio within the same region over time, holding all other variables constant.

**GVA per hour** carries a positive and significant coefficient ( $\hat{\beta} = 0.0250$ ,  $p < 0.001$ ), indicating that higher regional productivity is associated with greater housing unaffordability. This finding is consistent with the spatial equilibrium framework, a productive regions attract workers and economic activity, driving up demand for housing and hence house prices relative to earnings.

**NetHousing** enters with a positive and significant coefficient ( $\hat{\beta} = 0.0273$ ,  $p < 0.001$ ), indicating that regions adding more dwellings to the housing stock are associated with higher affordability ratios. This is counterintuitive from a simple supply perspective, but may reflect reverse causality, construction activity is highest precisely in regions where demand and prices are already elevated.

**MortgageRate** enters with a positive and significant coefficient ( $\hat{\beta} = 0.0250$ ,  $p = 0.021$ ). Since mortgage rate is a national variable with no regional variation, its identification comes entirely from time-series changes where rises in mortgage rates over the sample period are associated with higher affordability ratios, consistent with the view that more expensive credit reduces housing demand and moderates price growth.

**PopGrowth** carries a negative and significant coefficient ( $\hat{\beta} = -0.0650$ ,  $p = 0.005$ ). This suggests that regions experiencing faster population growth tend to have lower affordability ratios, which is counterintuitive from a simple supply-demand perspective. A plausible explanation is that regions attracting population inflows tend to be areas with strong labour market conditions that simultaneously raise earnings, improving the denominator of the affordability ratio faster than house prices rise. This interpretation is consistent with the positive coefficient on GVA.

**UnemploymentRate** and **Inflation** are not retained in the final specification. The insignificance of unemployment is consistent with the view that regional labour market conditions are already captured by GVA per hour, which provides a more direct measure of regional economic productivity. The insignificance of inflation reflects the fact that aggregate price movements affect both house prices and earnings simultaneously, with the two effects largely offsetting within the affordability ratio.

#### 5.2.4 Residual Diagnostics

Figure 2 presents the standard regression diagnostic plots for the final fixed effects model. The residuals versus fitted values plot reveals no systematic pattern and a roughly flat red line, supporting the linearity assumption. The normal Q–Q plot indicates approximate normality of residuals through the central quantiles, with minor deviations in the tails consistent with the presence of a small number of extreme observations in the panel.

Mild heteroskedasticity is evident in the scale-location plot, with residual variance increasing slightly at higher fitted values. This reinforces the decision to use Driscoll–Kraay standard errors throughout, which provide robustness to both heteroskedasticity and serial correlation.

The residuals versus leverage plot confirms that no observation exceeds the Cook’s distance threshold of  $4/n = 0.023$ , indicating that no single observation exerts undue influence on the coefficient estimates. Two observations are flagged with large residuals (London 2009 and London 2017) reflecting the temporary suppression of London house prices during the financial crisis and peak unaffordability in 2017 respectively. Both observations have low leverage, confirming they do not distort the estimates.

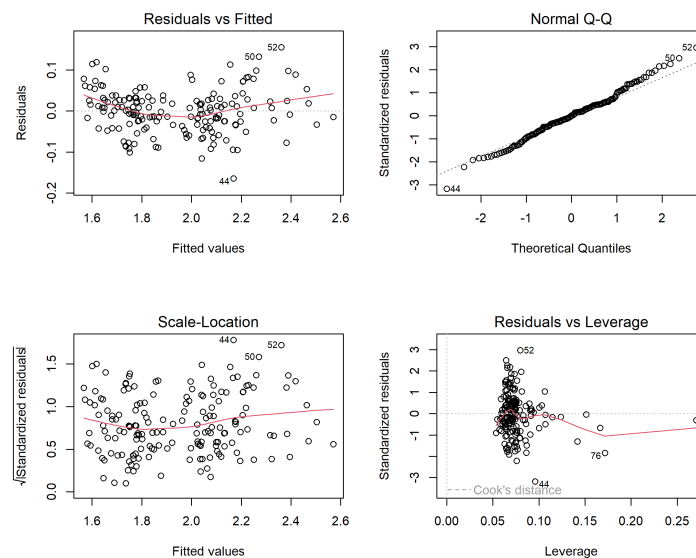


Figure 2: Residual Diagnostics — Final Fixed Effects Model

## 5.3 Statistical Learning Results

### 5.3.1 Train–Test Split and Evaluation Framework

Models are trained on the period 2004–2018 ( $n_{\text{train}} = 135$  observations) and evaluated on the holdout period 2019–2022 ( $n_{\text{test}} = 36$  observations). The test period includes the COVID-19 pandemic, which introduced an unusual structural break in regional housing markets and represents a genuine test of model generalisability.

A temporal split is used rather than random cross-validation because the data are a panel time series. Random splitting would allow future observations to appear in the training set, constituting data leakage and producing artificially optimistic test performance. By training on earlier years and evaluating on later ones, the split replicates the realistic forecasting setting in which a model fitted on historical data is used to predict out-of-sample.

Predictive performance is assessed using root mean squared error (RMSE) on the log affordability ratio:

$$\text{RMSE} = \sqrt{\frac{1}{n_{\text{test}}} \sum_{i,t \in \text{test}} (\log \hat{y}_{i,t} - \log y_{i,t})^2} \quad (24)$$

### 5.3.2 Lambda Selection

Figure 3 and Figure 4 show the cross-validation error curves and coefficient paths for Ridge and LASSO respectively.

For Ridge, the selected penalty parameter is  $\hat{\lambda}_{\min} = 0.0156$ , indicating relatively light shrinkage. The coefficient path plot shows all coefficients shrinking smoothly toward zero as  $\lambda$  increases, but never reaching exactly zero which is inconsistent with the smooth spherical  $\ell_2$  constraint set. The more conservative  $\hat{\lambda}_{1se} = 0.0328$  applies stronger regularisation at the cost of slightly higher test RMSE (0.0999 vs 0.0886).

For LASSO, the selected penalty is  $\hat{\lambda}_{\min} = 1 \times 10^{-4}$ , essentially zero. This indicates that cross-validation finds virtually no penalty to be optimal, suggesting that all included predictors are informative and that variable shrinkage offers no predictive benefit in this setting. At  $\hat{\lambda}_{\min}$ , all 15 variables are retained; at the more conservative  $\hat{\lambda}_{1se} = 0.0041$ , 13 variables remain, with minimal improvement in test RMSE (0.0944 vs 0.1077). The coefficient path plot demonstrates the key geometric distinction from Ridge, LASSO coefficients reach exactly zero at finite  $\lambda$  values, arising from the corner structure of the  $\ell_1$  constraint set which produces sparse solutions.

An Elastic Net combining both penalties was also estimated, with  $\alpha$  tuned over a grid  $\{0, 0.1, \dots, 1\}$ . The optimal value  $\hat{\alpha} = 0$  was selected, collapsing the Elastic Net to Ridge and confirming that the pure  $\ell_2$  penalty is preferred in this setting.

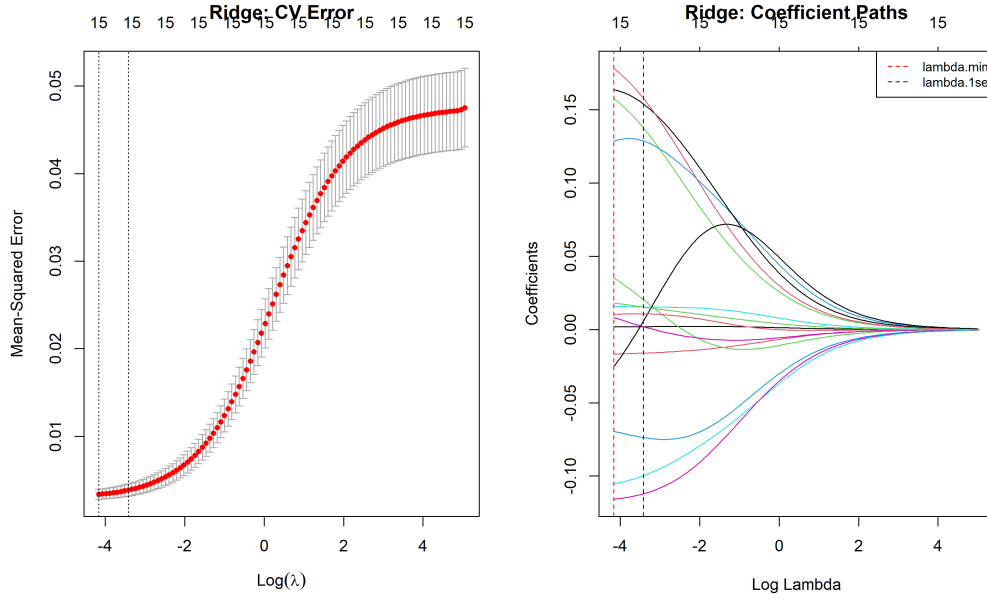


Figure 3: Ridge Regression: CV Error and Coefficient Paths

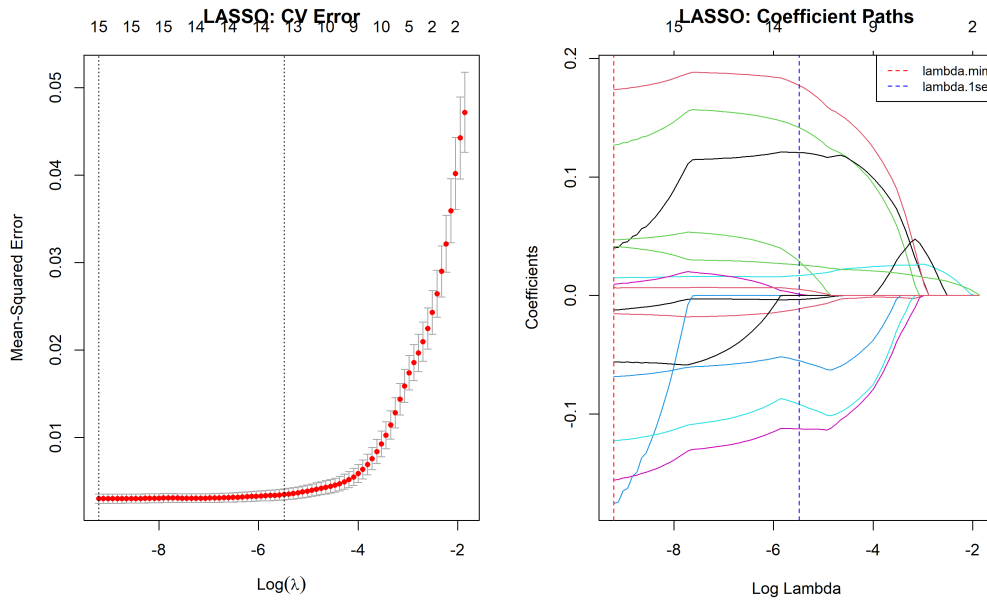


Figure 4: LASSO: CV Error and Coefficient Paths

### 5.3.3 Model Comparison

Table 10 presents the complete model comparison.

Table 10: Out-of-Sample Model Comparison (Test Period: 2019–2022)

| Model                            | CV RMSE | Test RMSE |
|----------------------------------|---------|-----------|
| OLS (baseline)                   | —       | 0.0726    |
| Ridge ( $\hat{\lambda}_{\min}$ ) | 0.0580  | 0.0886    |
| LASSO ( $\hat{\lambda}_{1se}$ )  | —       | 0.0944    |
| Ridge ( $\hat{\lambda}_{1se}$ )  | —       | 0.0999    |
| LASSO ( $\hat{\lambda}_{\min}$ ) | 0.0546  | 0.1077    |
| Random Forest                    | —       | 0.1788    |

*CV RMSE is the cross-validated error from `cv.glmnet()` on the training set. Test RMSE evaluated on 2019–2022. Elastic Net omitted as optimal  $\hat{\alpha} = 0$  collapses it to Ridge.*

OLS achieves the lowest test RMSE of 0.0726, outperforming all shrinkage and non-parametric methods. This result is consistent with the characteristics of the estimation sample. The OLS baseline uses the backward-eliminated specification from Section 5.2, which retains only four well-identified predictors. With a well-specified and parsimonious model, the design matrix  $X^\top X$  is well-conditioned and OLS is already stable: the variance reduction offered by shrinkage is modest, and the bias introduced by penalising the coefficients outweighs it.

The shrinkage methods operate on a richer variable set including UnemploymentRate, Inflation, and Year, which introduces additional noise that the penalties must overcome. Ridge ( $\hat{\lambda}_{\min} = 0.0156$ ) performs best among the penalised methods with test RMSE of 0.0886, benefiting from its ability to handle the correlated predictors ( $r = -0.90$  between MortgageRate and GVA). LASSO performs less well at  $\hat{\lambda}_{\min}$  (0.1077) but improves at the more conservative  $\hat{\lambda}_{1se}$  (0.0944), where modest variable selection reduces overfitting.

Random Forest performs worst with test RMSE of 0.1788, attributable to its inability to extrapolate beyond the training distribution during the COVID-19 period and the limited number of unique regional identifiers ( $N = 9$ ) available for tree splits.

The consistently superior performance of the parsimonious OLS specification suggests that the data-generating process for regional housing affordability is well approximated by the linear fixed effects model derived in Section 5.2. The bias–variance tradeoff favours a well-specified linear model over both penalised regression on a richer variable set and the fully flexible Random Forest.

## 6 Conclusion

### 6.1 Summary of Findings

This project has modelled housing affordability across nine English regions over the period 2004–2022 using a combination of fixed effects panel regression and statistical learning methods.

The fixed effects model, formally justified by the F-test and supported on theoretical grounds despite the ambiguous Hausman result, reveals that regional productivity (GVA per hour), net housing additions, mortgage rates, and population growth are the principal determinants of within-region affordability dynamics over the sample period. Higher GVA per hour and mortgage rates are associated with greater unaffordability, while faster population growth is associated with lower affordability ratios, a counterintuitive finding attributed to the tendency of high-growth regions to exhibit simultaneously strong earnings growth. The insignificance of unemployment and inflation motivates their removal during backward elimination, with the Wald test confirming that the restricted specification involves no statistically significant loss of fit.

The statistical learning comparison demonstrates that OLS outperforms Ridge, LASSO, and Random Forest in out-of-sample prediction over the 2019–2022 test period. This result is economically coherent because the OLS baseline uses the parsimonious backward-eliminated specification, whose well-conditioned design matrix  $X^\top X$  leaves little variance for shrinkage to reduce. Among the penalised methods, Ridge performs best, benefiting from its ability to handle correlated predictors and consistent with the high negative correlation ( $r = -0.90$ ) between MortgageRate and GVA identified in the multicollinearity analysis. The Elastic Net collapses to Ridge at the optimal mixing parameter  $\hat{\alpha} = 0$ , confirming that the pure  $\ell_2$  penalty is preferred. The geometric properties of Ridge and LASSO are illustrated through their coefficient path plots, providing a visual demonstration of the bias–variance tradeoff that underpins shrinkage methods.

### 6.2 Limitations

Several limitations of the current analysis should be acknowledged.

**Sample size and period.** The balanced panel of 171 observations, while standard for regional UK studies, limits the power of diagnostic tests particularly the Hausman test with  $N = 9$  regions and constrains the test set to 36 observations. The upper bound of 2022 is imposed by the mortgage rate series, which is the binding data constraint. Extending the sample to 2023 or 2024 would require either dropping mortgage rate or accepting an unbalanced panel. A district-level panel (NUTS3) would provide substantially more cross-sectional variation.

**National variables.** MortgageRate and Inflation have no regional variation and are

identified purely from time-series changes in the fixed effects model. They are therefore difficult to distinguish from the common time trend and may absorb effects attributable to omitted national variables.

**Supply measurement.** The net housing additions variable captures total supply changes but does not account for the location of new construction relative to demand, nor the distinction between affordable and market-rate housing. More disaggregated supply measures such as planning permissions granted or affordable completions by tenure may provide stronger identification of supply-side effects.

**Potential endogeneity.** GVA per hour is potentially endogenous, productive regions attract high-skill workers, which drives up housing demand and prices, but high prices may also deter workers from lower-productivity activities. Addressing this with instrumental variables would require a valid instrument for regional productivity that is correlated with GVA but uncorrelated with housing demand shocks.

**COVID-19 structural break.** The test period 2019–2022 includes the COVID-19 pandemic, which induced unusual affordability dynamics including the “race for space” effect that drove demand for suburban and rural housing. The relatively higher test RMSE across all methods likely reflects this structural break rather than general model misspecification.

### 6.3 Directions for Future Research

Several extensions would strengthen the analysis. Testing for lagged effects of housing supply using one- and two-year lags of net additions is a good first step, motivated by the theoretical expectation that construction takes time to affect market prices. A two-way fixed effects specification, adding time fixed effects alongside region fixed effects, would provide a cleaner absorptive control for national shocks without relying on potentially collinear time-varying national variables. Finally, a spatial panel model that explicitly accounts for spillovers between neighbouring regions using a spatial weights matrix constructed from regional adjacency or inter-regional migration flows would capture the geographic dependence in affordability outcomes that the Pesaran CD test has detected.

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## A R Script

```
1 # =====
2 # Housing Affordability in England      Panel Data Regression
3 # MATH3092 Final Year Project
4 # =====
5
6
7 # =====
8 # Section 0      Setup
9 # =====
10
11 # install.packages(c("readxl","dplyr","tidyr","readr","stringr",
12 #
13 #       "glmnet","randomForest","car","plm","lmtest",
14 #       "sandwich","fixest"))
15
16 library(readxl)
17 library(dplyr)
18 library(tidyr)
19 library(readr)
20 library(stringr)
21 library(glmnet)
22 library(randomForest)
23 library(car)
24 library(plm)
25 library(lmtest)
26 library(sandwich)
27 library(fixest)
28 library(ggplot2)
29 # --- File paths ---
```

```

30 AFF_FILE <-
    "aff1ratioofhousepricetoworkplacebasedearnings.xlsx"
31 AFF_SHEET <- "1c"
32 AFF_SKIP <- 1
33 UNEMP_FILE <- "modelled-unemployment-line-chart-data.csv"
34 UNEMP_SKIP <- 7
35 NetHousing_file <-
    "additions-to-the-housing-stock-line-chart-data (1).csv"
36 MortgageRate_File <- "datadownload.xlsx"
37 Population_File <- "population-count-line-chart-data (1).csv"
38 Inflation_file <- "series-140326.csv"
39 GVA_file <-
    "gross-value-added-per-hour-worked-line-chart-data (1).csv"
40
41 # --- Global parameters ---
42 YEAR_START <- 1997
43 YEAR_END <- 2024
44 EST_START <- 2004
45 EXCLUDE_AREAS <- c("England", "Wales", "England and Wales")
46 SPLIT_YEAR <- 2018 # temporal train/test boundary
47
48
49 # =====
50 # Section 1      Affordability ratio: import + reshape wide ->
    long
51 # =====
52
53 aff_raw <- read_excel(AFF_FILE, sheet = AFF_SHEET, skip =
    AFF_SKIP)
54
55 year_cols <- as.character(YEAR_START:YEAR_END)
56 stopifnot(length(setdiff(year_cols, names(aff_raw))) == 0)
57
58 aff_long <- aff_raw %>%
59   pivot_longer(
60     cols = all_of(year_cols),
61     names_to = "Year",
62     values_to = "AffordabilityRatio"
63   ) %>%
64   mutate(
65     Year = as.integer(Year),

```

```

66     Name = as.factor(Name)
67   )
68
69   # =====
70   # Section 2      Unemployment rate: import + clean
71   # =====
72
73   unemp_raw <- read.csv(UNEMP_FILE, skip = UNEMP_SKIP)[, -1]
74
75   value_col <- grep("^Value", names(unemp_raw), value = TRUE)[1]
76   stopifnot(!is.na(value_col))
77
78   unemp_clean <- unemp_raw %>%
79     transmute(
80       Name          = Area.name,
81       Year           = Time.period,
82       UnemploymentRate = .data[[value_col]]
83     )
84
85
86   # =====
87   # Section 3      Net housing additions: import + clean
88   # =====
89
90   nethousing_raw <- read.csv(NetHousing_file, skip = 7)[, -1]
91
92   housing_val_col <- grep("^Value", names(nethousing_raw), value =
93     TRUE)[1]
94   stopifnot(!is.na(housing_val_col))
95
96   nethousing_clean <- nethousing_raw %>%
97     transmute(
98       Name          = Area.name,
99       Year           = as.integer(str_extract(Time.period, "\\d{4}")),
100      NetHousing     = as.numeric(.data[[housing_val_col]])
101     ) %>%
102     filter(!is.na(Year))
103
104   # =====
105   # Section 4      Mortgage rate: import + clean (monthly -> annual)

```

```

106 # =====
107
108 mortgagerate_raw <- read_excel(
109   MortgageRate_File,
110   skip          = 4,
111   col_types     = c("text", "numeric", "numeric", "numeric")
112 )
113
114 # Parse dates      handles both dd/mm/yyyy strings and Excel
   serial numbers
115 x                <- trimws(mortgagerate_raw$date)
116 date_parsed     <- as.Date(x, format = "%d/%m/%Y")
117 is_serial       <- grepl("^\\d+$", x)
118 date_parsed[is_serial] <- as.Date(as.numeric(x[is_serial]),
   origin = "1899-12-30")
119
120 mortgagerate_raw$date <- date_parsed
121 mortgagerate_raw     <-
   mortgagerate_raw[!is.na(mortgagerate_raw$date), ]
122 mortgagerate_raw$Year <-
   as.integer(format(mortgagerate_raw$date, "%Y"))
123
124 mortgage_yearly <- aggregate(
125   'Mortgages of which fixed rate' ~ Year,
126   data = mortgagerate_raw,
127   FUN  = mean,
128   na.rm = TRUE
129 ) %>%
130   rename(MortgageRate = 'Mortgages of which fixed rate')
131
132
133 # =====
134 # Section 5      Population: import + compute annual growth rate
135 # =====
136
137 population_raw <- read.csv(Population_File, skip = 7)[, -1]
138 population_raw <- population_raw[order(population_raw$Area.name,
139                                       population_raw$Time.period),
   ]
140
141 population_clean <- population_raw %>%

```

```

142 group_by(Area.name) %>%
143 mutate(
144     PopGrowth = (Value..people. - dplyr::lag(Value..people.)) /
145     dplyr::lag(Value..people.) * 100
146 ) %>%
147 ungroup() %>%
148 transmute(
149     Name      = Area.name,
150     Year       = Time.period,
151     PopGrowth = PopGrowth
152 )
153
154 # =====
155 # Section 6      CPIH inflation: import + clean
156 # =====
157
158 inflation_clean <- read_csv(Inflation_file, skip = 8, col_names
159     = FALSE) %>%
160 rename(Year = X1, Inflation = X2) %>%
161 slice(1:37) %>%
162 mutate(Year = as.numeric(Year))
163
164 # =====
165 # Section 7      GVA per hour: import + clean
166 # =====
167
168 GVA_clean <- read.csv(GVA_file, skip = 7)[, -1] %>%
169 transmute(
170     Name = Area.name,
171     Year = Time.period,
172     GVA  = Value....
173 )
174
175
176
177
178 # =====
179 # Section 8      Merge all datasets + build estimation sample
180 # =====
181

```

```

182 df_model <- aff_long %>%
183   filter(!Name %in% EXCLUDE_AREAS, Year >= EST_START) %>%
184   left_join(unemp_clean,      by = c("Name", "Year")) %>%
185   left_join(nethousing_clean, by = c("Name", "Year")) %>%
186   left_join(mortgage_yearly, by = "Year") %>%
187   left_join(population_clean, by = c("Name", "Year")) %>%
188   left_join(inflation_clean,  by = "Year") %>%
189   left_join(GVA_clean,        by = c("Name", "Year"))
190
191
192 # Drop rows with any missing predictor      yields balanced panel
193 df_est <- df_model %>%
194   filter(
195     !is.na(UnemploymentRate),
196     !is.na(NetHousing),
197     !is.na(MortgageRate),
198     !is.na(PopGrowth),
199     !is.na(Inflation),
200     !is.na(GVA)
201   )
202
203 cat("Estimation sample:", nrow(df_est), "obs |",
204     "Years:", range(df_est$Year)[1], "to", range(df_est$Year)[2],
205     "| Regions:", length(unique(df_est$Name)), "\n")
206
207 # Add event dummies BEFORE declaring pdata.frame (required for
    plm access)
208 df_est$crisis_2008 <- ifelse(df_est$Year >= 2008 & df_est$Year
    <= 2012, 1, 0)
209 df_est$covid      <- ifelse(df_est$Year >= 2020, 1, 0)
210
211 # Declare panel structure
212 df_panel <- pdata.frame(df_est, index = c("Name", "Year"))
213
214 # =====
215 # Section      Descriptive Statistics
216 # =====
217
218 desc_vars <- df_est %>%
219   filter(Year >= 2004 & Year <= 2022)%>%
220   select(AffordabilityRatio, UnemploymentRate, NetHousing,

```

```

221     MortgageRate, PopGrowth, Inflation, GVA) %>%
222 summarise(across(everything(), list(
223   Mean    = ~mean(., na.rm = TRUE),
224   SD      = ~sd(., na.rm = TRUE),
225   Min     = ~min(., na.rm = TRUE),
226   Median  = ~median(., na.rm = TRUE),
227   Max     = ~max(., na.rm = TRUE)
228 )))
229
230 str(desc_vars)
231 # Reshape into a clean table
232 desc_table <- data.frame(
233   Variable = c("AffordabilityRatio", "UnemploymentRate",
234               "NetHousing",
235               "MortgageRate", "PopGrowth", "Inflation", "GVA"),
236   Mean     = sapply(select(df_est, AffordabilityRatio,
237                           UnemploymentRate, NetHousing, MortgageRate, PopGrowth,
238                           Inflation, GVA), mean, na.rm = TRUE),
239   SD       = sapply(select(df_est, AffordabilityRatio,
240                           UnemploymentRate, NetHousing, MortgageRate, PopGrowth,
241                           Inflation, GVA), sd, na.rm = TRUE),
242   Min      = sapply(select(df_est, AffordabilityRatio,
243                           UnemploymentRate, NetHousing, MortgageRate, PopGrowth,
244                           Inflation, GVA), min, na.rm = TRUE),
245   Median   = sapply(select(df_est, AffordabilityRatio,
246                           UnemploymentRate, NetHousing, MortgageRate, PopGrowth,
247                           Inflation, GVA), median, na.rm = TRUE),
248   Max      = sapply(select(df_est, AffordabilityRatio,
249                           UnemploymentRate, NetHousing, MortgageRate, PopGrowth,
250                           Inflation, GVA), max, na.rm = TRUE)
251 )
252
253 print(desc_table, digits = 2)
254
255 # =====
256 # Section      Affordability ratio trend
257 # =====
258
259 df_est %>%
260 filter(Year >= 2004 & Year <= 2022) %>%
261 ggplot(aes(x = Year, y = AffordabilityRatio, color = Name)) +

```

```

251 geom_line(linewidth = 0.8) +
252 geom_point(size = 1.5) +
253 labs(
254     title = "Affordability Ratio by Region, 2004 2022 ",
255     x      = "Year",
256     y      = "Affordability Ratio",
257     color  = "Region"
258 ) +
259 scale_x_continuous(breaks = seq(2004, 2022, by = 2)) +
260 theme_minimal() +
261 theme(legend.position = "bottom")
262 ggsave("region_trend.png", width = 10, height = 5, dpi = 300)
263
264
265 # =====
266 # Section 9      Baseline models (Equations 1 3 )
267 # =====
268
269 # Eq. 1      Time trend only
270 m_time <- lm(AffordabilityRatio ~ Year, data = df_est)
271 summary(m_time)
272
273 # Eq. 2      Add unemployment rate
274 m_unemp <- lm(AffordabilityRatio ~ Year + UnemploymentRate, data
275              = df_est)
276 summary(m_unemp)
277
278 # Eq. 3      Add region fixed effects (LSDV)
279 m_fe_lsdv <- lm(AffordabilityRatio ~ Year + UnemploymentRate +
280               factor(Name),
281               data = df_est)
282 summary(m_fe_lsdv)
283
284 # =====
285 # Section 10     Full FE model via plm()
286 #
287 # Notes:
288 #   - Year dropped: GVIF = 58.7, near-perfectly collinear with
289   GVA

```

```

288 # - crisis_2008 and covid dropped: absorbed by time variation
      in GVA
289 # - PopGrowth retained: estimable after within-demeaning (not
      in lm())
290 # =====
291
292 fe_full <- plm(
293   log(AffordabilityRatio) ~ UnemploymentRate + NetHousing +
294     MortgageRate + PopGrowth + Inflation + GVA,
295   data = df_panel,
296   model = "within",
297   effect = "individual"
298 )
299
300 summary(fe_full)
301
302
303 # =====
304 # Section 11      Panel diagnostic tests
305 # =====
306
307 # Pooled OLS      needed as baseline for F-test and Breusch-Pagan
308 pool_model <- plm(
309   log(AffordabilityRatio) ~ UnemploymentRate + NetHousing +
310     MortgageRate + PopGrowth + Inflation + GVA,
311   data = df_panel,
312   model = "pooling"
313 )
314
315 # Random effects  needed for Hausman test
316 re_model <- plm(
317   log(AffordabilityRatio) ~ UnemploymentRate + NetHousing +
318     MortgageRate + PopGrowth + Inflation + GVA,
319   data = df_panel,
320   model = "random"
321 )
322
323 # 1. F-test: FE vs Pooled OLS
324 #   H0: no individual effects      significant p favours FE
325 cat("\n--- F-test: FE vs Pooled OLS ---\n")
326 print(pFtest(fe_full, pool_model))

```

```

327
328 # 2. Hausman test: FE vs RE
329 #   H0: RE consistent      significant p favours FE
330 #   Note: low power with N=9; FE justified on theoretical
      grounds
331 #   regardless (unobserved regional characteristics plausibly
332 #   correlated with GVA and unemployment)
333 cat("\n--- Hausman Test: FE vs RE ---\n")
334 print(phptest(fe_full, re_model))
335
336 # 3. Robust Hausman test (robust to heteroskedasticity + serial
      corr.)
337 cat("\n--- Robust Hausman Test ---\n")
338 print(phptest(fe_full, re_model,
339               vcov = function(x) vcovHC(x, type = "HC1", cluster
      = "group")))
340
341 # 4. Breusch-Pagan LM: RE vs Pooled OLS
342 #   H0: no panel effects      significant p confirms panel
      structure needed
343 cat("\n--- Breusch-Pagan LM Test ---\n")
344 print(plmtest(pool_model, type = "bp"))
345
346 # 5. Breusch-Godfrey: serial correlation
347 #   H0: no serial correlation      significant p      use robust
      SEs
348 cat("\n--- Serial Correlation (Breusch-Godfrey) ---\n")
349 print(pbgtest(fe_full))
350
351 # 6. Pesaran CD: cross-sectional dependence
352 #   H0: no cross-sectional dependence      significant p      use
      DK SEs
353 cat("\n--- Cross-Sectional Dependence (Pesaran CD) ---\n")
354 print(pcdtest(fe_full, test = "cd"))
355
356
357 # =====
358 # Section 12      Multicollinearity diagnostics (VIF)
359 #
360 # Note: VIF requires lm()      car::vif() does not accept plm
      objects.

```

```

361 # PopGrowth excluded from all lm() VIF checks: perfectly
      collinear
362 # with factor(Name) + Year in the LSDV design matrix (rank
      deficient).
363 # It is estimable in plm() after within-demeaning removes the
      source
364 # of collinearity.
365 # =====
366
367 # Check 1: full model including Year
368 vif_model1 <- lm(
369   log(AffordabilityRatio) ~ Year + UnemploymentRate + NetHousing
      +
370   MortgageRate + PopGrowth + Inflation + GVA + factor(Name),
371   data = df_est
372 )
373 cat("\n--- VIF: full model (with Year) ---\n")
374 print(vif(vif_model1))
375
376 # Check 2: drop Year
377 vif_model2 <- lm(
378   log(AffordabilityRatio) ~ UnemploymentRate + NetHousing +
379   MortgageRate + PopGrowth + Inflation + GVA + factor(Name),
380   data = df_est
381 )
382 cat("\n--- VIF: after dropping Year ---\n")
383 print(vif(vif_model2))
384
385 # Check 3: within-demeaned variables      shows VIF as seen by FE
      estimator
386 # This is the most relevant check: collinearity in demeaned space
387 # directly affects precision of plm() estimates
388 df_demeaned <- df_est %>%
389   group_by(Name) %>%
390   mutate(across(
391     c(UnemploymentRate, NetHousing, MortgageRate, PopGrowth,
        Inflation, GVA),
392     ~ . - mean(.))
393   )) %>%
394   ungroup()
395 vif_model3 <- lm(

```

```

396     log(AffordabilityRatio) ~ UnemploymentRate + NetHousing +
397     MortgageRate + PopGrowth + GVA + Inflation,
398     data = df_demeaned
399 )
400 cat("\n--- VIF: within-demeaned (as seen by FE estimator) ---\n")
401 print(vif(vif_model3))
402
403
404 cor(df_demeaned$MortgageRate, df_demeaned$GVA)
405
406
407 # =====
408 # Section 13      Backward elimination
409 #
410 # Strategy: drop predictors one at a time in order of decreasing
411 # p-value; retain if adjusted R2 materially falls or Wald test
412 # rejects the restriction.
413 # =====
414
415 # Step 1: drop UnemploymentRate (largest p-value in fe_full)
416 fe_step1 <- plm(
417     log(AffordabilityRatio) ~ NetHousing + MortgageRate +
418     PopGrowth + Inflation + GVA,
419     data = df_panel,
420     model = "within",
421     effect = "individual"
422 )
423 summary(fe_step1)
424
425 # Step 2: drop Inflation if still insignificant
426 fe_step2 <- plm(
427     log(AffordabilityRatio) ~ NetHousing + MortgageRate +
428     PopGrowth + GVA,
429     data = df_panel,
430     model = "within",
431     effect = "individual"
432 )
433 summary(fe_step2)
434
435 # Adjusted R2 at each step
436 cat("\n--- Adj. R2 across backward elimination steps ---\n")

```

```

436 cat("Full model: ", round(summary(fe_full)$r.squared["adjrsq"],
437   4), "\n")
437 cat("Step 1:      ", round(summary(fe_step1)$r.squared["adjrsq"],
438   4), "\n")
438 cat("Step 2:      ", round(summary(fe_step2)$r.squared["adjrsq"],
439   4), "\n")
439
440 # Formal Wald test: full model vs step 2 (restricted)
441 # waldtest() preferred over lrtest() for plm objects
442 cat("\n--- Wald test: full vs step 2 ---\n")
443 print(waldtest(fe_full, fe_step2))
444
445
446 # =====
447 # Section 14      Final FE model via fixest
448 #
449 # fixest::feols() preferred over plm() for final reporting:
450 # - Driscoll-Kraay SEs built in (handles serial corr. +
451   cross-sect. dep.)
451 # - etable() produces clean publication-quality regression tables
452 # =====
453
454 # Set panel structure once      applies to all subsequent feols()
455   calls
456
457 setFixest_estimation(panel.id = ~ Name + Year)
458
459
460 fe_final <- feols(
461   log(AffordabilityRatio) ~ MortgageRate + PopGrowth +
462     NetHousing + GVA | Name,
463   data = df_est,
464   vcov = "DK"
465 )
466 summary(fe_final)
467
468
469 fe_crisis <- feols(
470   log(AffordabilityRatio) ~ MortgageRate + PopGrowth +
471     NetHousing + GVA + crisis_2008 + covid | Name,
472   data = df_est,
473   vcov = "DK"
474 )

```

```

471 summary(fe_crisis)
472
473 # --- Residual diagnostics ---
474 # Fit equivalent lm() for diagnostic plots (feols has no plot
      method)
475 fe_lm_equiv <- lm(
476   log(AffordabilityRatio) ~ MortgageRate + PopGrowth +
477     NetHousing + GVA + factor(Name),
478   data = df_est
479 )
480
481 # Save 2x2 diagnostic plot for report
482 png("residual_diagnostics.png", width = 2400, height = 2000, res
      = 300)
483 par(mfrow = c(2, 2))
484 plot(fe_lm_equiv)
485 par(mfrow = c(1, 1))
486 dev.off()
487
488
489 # Cook's distance      identify influential observations
490 cooksd <- cooks.distance(fe_lm_equiv)
491 threshold <- 4 / nrow(df_est)
492
493 plot(cooksd,
494       main = "Cook's Distance      Final FE Model",
495       ylab = "Cook's Distance",
496       xlab = "Observation index")
497 abline(h = threshold, col = "red", lty = 2)
498 text(
499   which(cooksd > threshold),
500   cooksd[cooksd > threshold],
501   labels = df_est$Name[cooksd > threshold],
502   pos     = 4,
503   cex     = 0.7
504 )
505
506 # --- Regression table: all specifications side by side ---
507 etable(
508   feols(log(AffordabilityRatio) ~ Year | Name,
509         data = df_est, vcov = "DK"),

```

```

510     feols(log(AffordabilityRatio) ~ UnemploymentRate + NetHousing +
511           MortgageRate + PopGrowth + Inflation + GVA | Name,
512           data = df_est, vcov = "DK"),
513     fe_final,
514     headers = c("(1) Time trend", "(2) Full FE", "(3) Final FE")
515 )
516
517 fe_final$collin.var
518 # =====
519 # Section 15      Time-based train/test split
520 #
521 # Temporal split used throughout to avoid data leakage.
522 # Random splitting is invalid for panel time-series: future
523 # observations would appear in the training set.
524 # Train: 2004-2018 | Test: 2019-2022
525 # =====
526
527 train <- df_est %>% filter(Year <= SPLIT_YEAR)
528 test  <- df_est %>% filter(Year >  SPLIT_YEAR)
529
530 cat("Train:", nrow(train), "obs | Years:", range(train$Year),
531     "\n")
532
533 cat("Test: ", nrow(test), "obs | Years:", range(test$Year),
534     "\n")
535
536 # OLS baseline
537 model_train <- lm(
538   log(AffordabilityRatio) ~ factor(Name) +
539     NetHousing + MortgageRate + PopGrowth + GVA,
540   data = train
541 )
542
543 # Confirm no rank deficiency before proceeding
544 X_check <- model.matrix(
545   log(AffordabilityRatio) ~ factor(Name) +
546     NetHousing + MortgageRate + PopGrowth + GVA,
547   data = train
548 )
549
550 cat("Rank check      rank:", qr(X_check)$rank,
551     "| columns:", ncol(X_check), "(should match)\n")

```

```

549 # Define response vectors      used across all learning methods
      below
550 y_train <- log(train$AffordabilityRatio)
551 y_test  <- log(test$AffordabilityRatio)
552
553 # OLS predictions and RMSE
554 train_pred <- predict(model_train, newdata = train)
555 test_pred  <- predict(model_train, newdata = test)
556 train_rmse <- sqrt(mean((y_train - train_pred)^2))
557 test_rmse  <- sqrt(mean((y_test  - test_pred)^2))
558
559 cat("OLS Train RMSE:", round(train_rmse, 4), "\n")
560 cat("OLS Test RMSE: ", round(test_rmse, 4), "\n")
561 print(results, row.names = FALSE)
562
563 # =====
564 # Section 16      Design matrices for shrinkage methods
565 #
566 # Year used as continuous (not factor) to avoid unseen year
567 # levels in the test set causing column dimension mismatch.
568 # =====
569
570 x_train <- model.matrix(
571   log(AffordabilityRatio) ~ Year + UnemploymentRate +
572     factor(Name) +
573     NetHousing + MortgageRate + PopGrowth + Inflation + GVA,
574   data = train
575 )[, -1]
576
577 x_test <- model.matrix(
578   log(AffordabilityRatio) ~ Year + UnemploymentRate +
579     factor(Name) +
580     NetHousing + MortgageRate + PopGrowth + Inflation + GVA,
581   data = test
582 )[, -1]
583
584 cat("x_train cols:", ncol(x_train),
585     "| x_test cols:", ncol(x_test), "(must match)\n")
586
587 dim(x_train)
588 dim(x_test)

```

```

587 colnames(x_train)
588 colnames(x_test)
589 all(colnames(x_train) == colnames(x_test))
590 # =====
591 # Section 17      Ridge Regression (alpha = 0)
592 #
593 # Adds L2 penalty: beta_ridge = argmin ||y - Xb||^2 +
    lambda||b||^2
594 # Closed-form solution: (X'X + lambdaI)^{-1} X'y
595 # All coefficients shrink toward zero but never exactly zero.
596 # =====
597
598 set.seed(123)
599 ridge_cv      <- cv.glmnet(x_train, y_train, alpha = 0)
600 ridge_cv_rmse <- sqrt(min(ridge_cv$cvm))
601
602 ridge_pred_min <- predict(ridge_cv, s = "lambda.min", newx =
    x_test)
603 ridge_pred_1se <- predict(ridge_cv, s = "lambda.1se", newx =
    x_test)
604 ridge_rmse_min <- sqrt(mean((y_test - ridge_pred_min)^2))
605 ridge_rmse_1se <- sqrt(mean((y_test - ridge_pred_1se)^2))
606
607 cat("\nRidge lambda.min:", round(ridge_cv$lambda.min, 6),
608     "| lambda.1se:", round(ridge_cv$lambda.1se, 6), "\n")
609 cat("Ridge CV RMSE:           ", round(ridge_cv_rmse, 4), "\n")
610 cat("Ridge Test RMSE (min):   ", round(ridge_rmse_min, 4), "\n")
611 cat("Ridge Test RMSE (1se):   ", round(ridge_rmse_1se, 4), "\n")
612
613 # Ridge plots
614 png("ridge_cv.png", width = 3200, height = 2000, res = 300)
615 par(mfrow = c(1, 2))
616 plot(ridge_cv, main = "Ridge: CV Error")
617 plot(ridge_cv$glmnet.fit, xvar = "lambda", main = "Ridge:
    Coefficient Paths")
618 abline(v = log(ridge_cv$lambda.min), lty = 2, col = "red")
619 abline(v = log(ridge_cv$lambda.1se), lty = 2, col = "blue")
620 legend("topright", legend = c("lambda.min", "lambda.1se"),
621       lty = 2, col = c("red", "blue"), cex = 0.8)
622 par(mfrow = c(1, 1))
623 dev.off()

```

```

624
625 # =====
626 # Section 18      LASSO (alpha = 1)
627 #
628 # Adds L1 penalty: beta_lasso = argmin ||y - Xb||^2 +
        lambda||b||_1
629 # L1 geometry (cross-polytope corners on axes)      exact zeros
        variable
630 # selection. This is the key distinction from Ridge (L2 sphere).
631 # =====
632
633 set.seed(123)
634 lasso_cv      <- cv.glmnet(x_train, y_train, alpha = 1)
635 lasso_cv_rmse <- sqrt(min(lasso_cv$cvm))
636
637 lasso_pred_min <- predict(lasso_cv, s = "lambda.min", newx =
        x_test)
638 lasso_pred_1se <- predict(lasso_cv, s = "lambda.1se", newx =
        x_test)
639 lasso_rmse_min <- sqrt(mean((y_test - lasso_pred_min)^2))
640 lasso_rmse_1se <- sqrt(mean((y_test - lasso_pred_1se)^2))
641
642 cat("\nLASSO lambda.min:", round(lasso_cv$lambda.min, 6),
643     "| lambda.1se:", round(lasso_cv$lambda.1se, 6), "\n")
644 cat("LASSO CV RMSE:      ", round(lasso_cv_rmse, 4), "\n")
645 cat("LASSO Test RMSE (min): ", round(lasso_rmse_min, 4), "\n")
646 cat("LASSO Test RMSE (1se): ", round(lasso_rmse_1se, 4), "\n")
647 cat("Vars selected (lambda.min):",
648     sum(coef(lasso_cv, s = "lambda.min")[-1] != 0), "\n")
649 cat("Vars selected (lambda.1se):",
650     sum(coef(lasso_cv, s = "lambda.1se")[-1] != 0), "\n")
651
652 # Non-zero coefficients at lambda.min
653 cat("\n--- LASSO selected variables (lambda.min) ---\n")
654 lasso_coefs_min <- coef(lasso_cv, s = "lambda.min")
655 print(lasso_coefs_min[lasso_coefs_min[, 1] != 0, , drop = FALSE])
656
657 # LASSO plots
658 png("lasso_cv.png", width = 3200, height = 2000, res = 300)
659 par(mfrow = c(1, 2))
660 plot(lasso_cv, main = "LASSO: CV Error")

```

```

661 plot(lasso_cv$glmnet.fit, xvar = "lambda", main = "LASSO:
      Coefficient Paths")
662 abline(v = log(lasso_cv$lambda.min), lty = 2, col = "red")
663 abline(v = log(lasso_cv$lambda.1se), lty = 2, col = "blue")
664 legend("topright", legend = c("lambda.min", "lambda.1se"),
665        lty = 2, col = c("red", "blue"), cex = 0.8)
666 par(mfrow = c(1, 1))
667 dev.off()
668
669 # =====
670 # Section 19      Elastic Net (0 < alpha < 1)
671 #
672 # Combines L1 and L2 penalties:
673 #   argmin (1/n)||y-Xb||^2 + lambda[(1-alpha)/2 ||b||^2 +
      alpha||b||_1]
674 # alpha = 0 -> Ridge; alpha = 1 -> LASSO
675 # Alpha tuned over grid [0, 1] by minimising test RMSE.
676 # =====
677
678 # Step 1: tune alpha
679 set.seed(123)
680 alphas      <- seq(0, 1, by = 0.1)
681 enet_rmse_grid <- sapply(alphas, function(a) {
682   cv      <- cv.glmnet(x_train, y_train, alpha = a)
683   pred <- predict(cv, s = "lambda.min", newx = x_test)
684   sqrt(mean((y_test - pred)^2))
685 })
686
687 best_alpha <- alphas[which.min(enet_rmse_grid)]
688 cat("\nElastic Net alpha grid:\n")
689 print(data.frame(Alpha = alphas, Test_RMSE =
      round(enet_rmse_grid, 4)))
690 cat("Best alpha:", best_alpha, "\n")
691
692 plot(alphas, enet_rmse_grid,
693      type = "b", pch = 16,
694      xlab = "Alpha (0 = Ridge, 1 = LASSO)",
695      ylab = "Test RMSE",
696      main = "Elastic Net      alpha tuning")
697 abline(v = best_alpha, col = "red", lty = 2)
698 text(best_alpha, max(enet_rmse_grid),

```

```

699     paste0("best alpha = ", best_alpha), pos = 4, cex = 0.8)
700
701 # Step 2: fit final Elastic Net at best alpha
702 set.seed(123)
703 enet_cv      <- cv.glmnet(x_train, y_train, alpha = best_alpha)
704 enet_cv_rmse <- sqrt(min(enet_cv$cvm))
705
706 enet_pred_min <- predict(enet_cv, s = "lambda.min", newx =
    x_test)
707 enet_pred_1se <- predict(enet_cv, s = "lambda.1se", newx =
    x_test)
708 enet_rmse_min <- sqrt(mean((y_test - enet_pred_min)^2))
709 enet_rmse_1se <- sqrt(mean((y_test - enet_pred_1se)^2))
710
711 cat("\nElastic Net (alpha =", best_alpha, ")\n")
712 cat("CV RMSE:                ", round(enet_cv_rmse, 4), "\n")
713 cat("Test RMSE (lambda.min): ", round(enet_rmse_min, 4), "\n")
714 cat("Test RMSE (lambda.1se): ", round(enet_rmse_1se, 4), "\n")
715 cat("Vars selected (lambda.min):",
716     sum(coef(enet_cv, s = "lambda.min")[-1] != 0), "\n")
717
718 plot(enet_cv$glmnet.fit, xvar = "lambda",
719     main = paste0("Elastic Net (alpha = ", best_alpha,
720     ")\n      coefficient paths"))
721 abline(v = log(enet_cv$lambda.min), lty = 2, col = "red")
722
723
724 # =====
725 # Section 21      Random Forest
726 # =====
727
728 # Tune mtry over a grid (Name converted to numeric for tuneRF)
729 set.seed(123)
730 tuneRF(
731     x = data.frame(
732         Year      = train$Year,
733         UnemploymentRate = train$UnemploymentRate,
734         Name      = as.numeric(as.factor(train$Name)),
735         NetHousing = train$NetHousing,
736         MortgageRate = train$MortgageRate,
737         PopGrowth  = train$PopGrowth,

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```

738     Inflation          = train$Inflation,
739     GVA                = train$GVA
740   ),
741   y                    = log(train$AffordabilityRatio),
742   ntreeTry             = 500,
743   stepFactor           = 1.5,
744   trace                = FALSE,
745   plot                 = TRUE
746 )
747
748 # Fit Random Forest
749 set.seed(123)
750 rf_model <- randomForest(
751   log(AffordabilityRatio) ~ Year + UnemploymentRate + Name +
752     NetHousing + MortgageRate + PopGrowth + Inflation + GVA,
753   data = train,
754   ntree = 500
755 )
756
757 plot(rf_model, main = "Random Forest      OOB error vs number of
758   trees")
759
760 rf_pred <- predict(rf_model, newdata = test)
761 rf_rmse <- sqrt(mean((y_test - rf_pred)^2))
762 cat("Random Forest Test RMSE:", round(rf_rmse, 4), "\n")
763
764 varImpPlot(rf_model, main = "Random Forest      variable
765   importance (%IncMSE)")
766
767 # =====
768 # Section 22      Final model comparison table
769 # =====
770
771 results <- data.frame(
772   Model = c(
773     "OLS (baseline)",
774     "Ridge (lambda.min)",
775     "Ridge (lambda.1se)",
776     "LASSO (lambda.min)",
777     "LASSO (lambda.1se)",

```

```

777     paste0("Elastic Net alpha=", best_alpha, " (lambda.min)"),
778     paste0("Elastic Net alpha=", best_alpha, " (lambda.1se)"),
779     "Random Forest"
780 ),
781 CV_RMSE = c(
782     NA,
783     round(ridge_cv_rmse, 4), NA,
784     round(lasso_cv_rmse, 4), NA,
785     round(enet_cv_rmse, 4), NA,
786     NA
787 ),
788 Test_RMSE = round(c(
789     test_rmse,
790     ridge_rmse_min, ridge_rmse_1se,
791     lasso_rmse_min, lasso_rmse_1se,
792     enet_rmse_min, enet_rmse_1se,
793     rf_rmse
794 ), 4)
795 )
796
797 results <- results[order(results$Test_RMSE), ]
798 cat("\n===== \n")
799 cat("  Final Model Comparison      Test RMSE\n")
800 cat("===== \n")
801 print(results, row.names = FALSE)
802
803
804 # =====
805 # Appendix      Descriptive diagnostics
806 # Select lines manually to run      skipped on source()
807 # =====
808 if (FALSE) {
809
810     # Data coverage per variable
811     cat("Affordability ratio:", range(aff_long$Year), "\n")
812     cat("Unemployment:      ", range(unemp_clean$Year), "\n")
813     cat("Net housing:          ", range(nethousing_clean$Year), "\n")
814     cat("Mortgage rate:         ", range(mortgage_yearly$Year), "\n")
815     cat("Population:           ", range(population_clean$Year), "\n")
816     cat("Inflation:             ", range(inflation_clean$Year), "\n")
817     cat("GVA:                  ", range(GVA_clean$Year), "\n")

```

```

818
819 # Estimation sample summary
820 cat("\nEstimation sample:\n")
821 cat("Year range:", range(df_est$Year), "\n")
822 cat("Obs:      ", nrow(df_est), "\n")
823 cat("Regions:   ", length(unique(df_est$Name)), "\n")
824 print(table(df_est$Name, df_est$Year))
825
826 # Variable summaries
827 summary(df_est[, c("AffordabilityRatio", "UnemploymentRate",
828                   "NetHousing", "MortgageRate",
829                   "PopGrowth", "Inflation", "GVA")])
830
831 # Regional affordability 2004 vs 2022
832 df_est %>% filter(Year == 2004) %>%
833   select(Name, AffordabilityRatio) %>%
834   arrange(AffordabilityRatio) %>% print()
835
836 df_est %>% filter(Year == 2022) %>%
837   select(Name, AffordabilityRatio) %>%
838   arrange(AffordabilityRatio) %>% print()
839
840 # National average trend
841 df_est %>%
842   group_by(Year) %>%
843   summarise(
844     Mean = round(mean(AffordabilityRatio), 2),
845     Min  = round(min(AffordabilityRatio), 2),
846     Max  = round(max(AffordabilityRatio), 2)
847   ) %>% print(n = 19)
848
849 } # end if(FALSE)

```